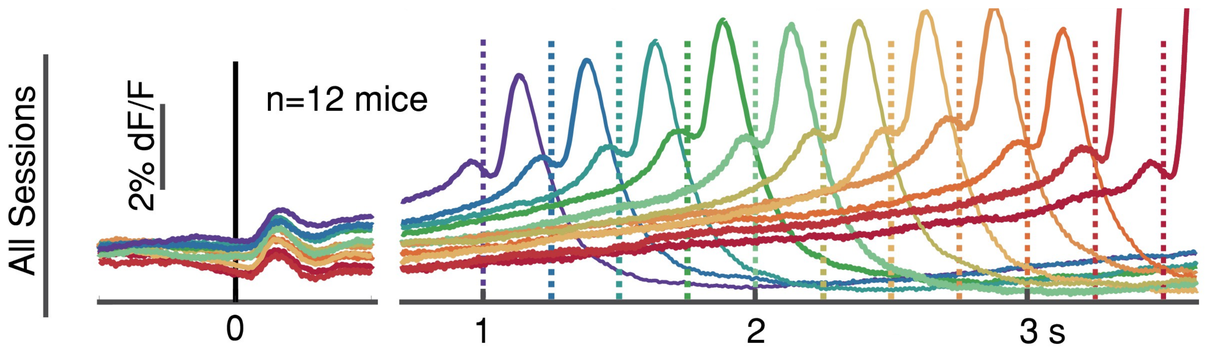
Neuromodulated Recurrent Neural Networks

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Objectives

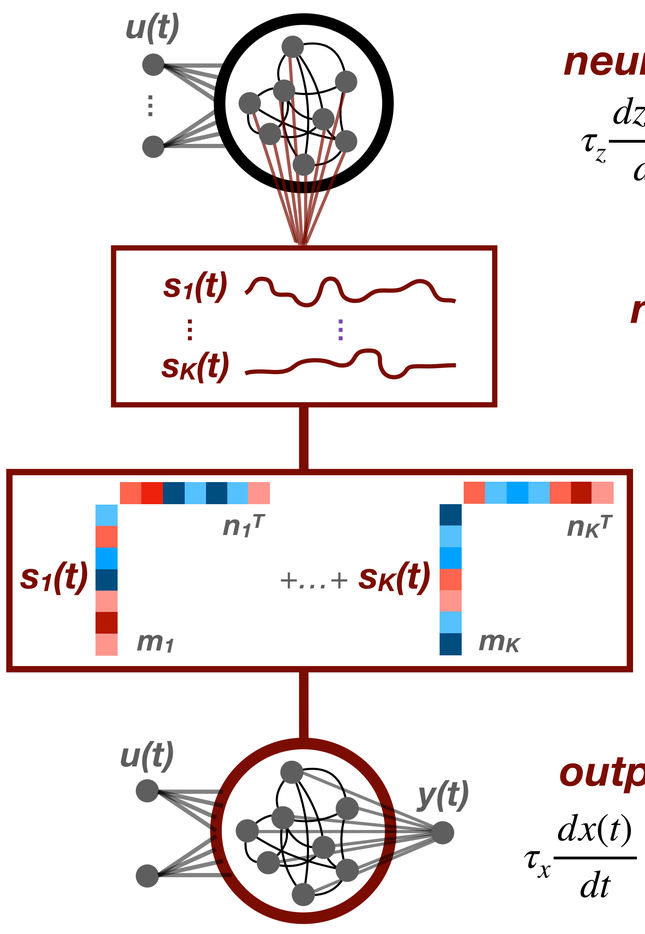


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- Neuromodulation is clearly important to brain function (e.g. dopamine in self-timed movements [1]), but it's uncommon in standard RNN models of neural circuits
- **Goal**: incorporate biologically plausible mechanism for neuromodulation in RNN models

Methods

Neuromodulated Low-Rank RNN



neuromodulatory subnetwork

$$\tau_z \frac{dz(t)}{dt} = -z(t) + \mathbf{W}_z \phi\left(z(t)\right) + \mathbf{B}_z u(t)$$

time-varying neuromodulatory signal

$$s\left(z(t)\right) = \sigma\left(\mathbf{A}_z z(t) + b_z\right)$$

neuromodulated low-rank weights

$$\mathbf{W}_x\left(z(t)\right) = \sum_{i=1}^{K} s_i(t) m_i n_i^\top$$

low-rank output-generating subnetwork

$$\tau_x \frac{dx(t)}{dt} = -x(t) + \mathbf{W}_x\left(z(t)\right)\phi\left(x(t)\right) + \mathbf{B}_x u(t)$$

$$y(t) = \mathbf{C}_x x(t)$$

- Model comprised of neuromodulatory and output-generating subnetworks
- Neuromodulatory signal modulates output network's low-rank recurrent weights

Intuition: Time Constants of Dynamics Modes

Assuming a linear, symmetric-weight output network with orthogonal m_i, we can reparameterize the system as a diagonal one with state $h(t) = \mathbf{M}^\top x(t)$…

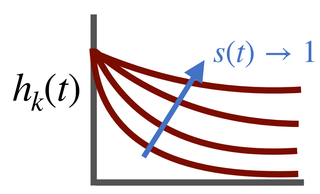
$$\tau_x \frac{dh(t)}{dt} = -h(t) + \mathbf{S}(t)h(t) \qquad h_k(t) = h_k(0)\exp\left(-\int_0^t \frac{\left(1 - s_k(t')\right)}{\tau_x} dt'\right)$$

$$\mathbf{S}(t) = \mathrm{diag}\left(s(t)\right)$$

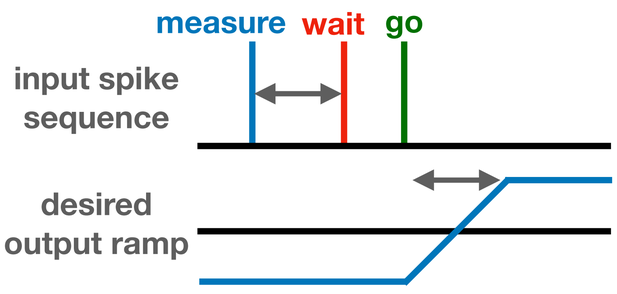
$h(t)$, in the column space of **M**, is controlled by the NM signal

$$\tau_x \frac{dh_\perp(t)}{dt} = -h_\perp(t) \qquad h_{\perp_k}(t) = h_{\perp_k}(0)\exp\left(-\frac{t}{\tau_x}\right)$$

$h_\perp(t)$, outside the column space, decays according to τ_x



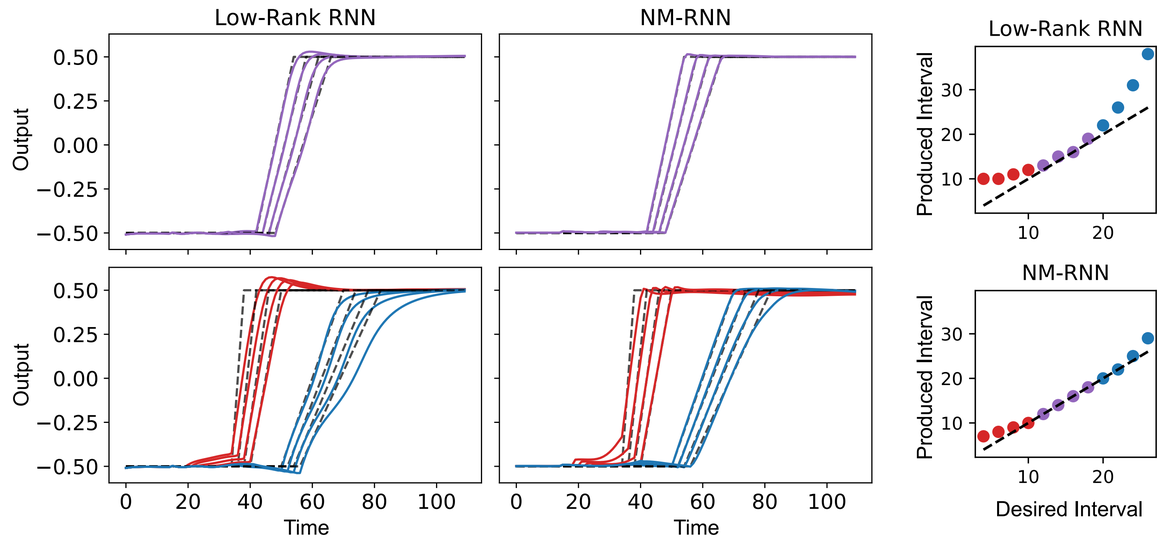
Measure-Wait-Go Task



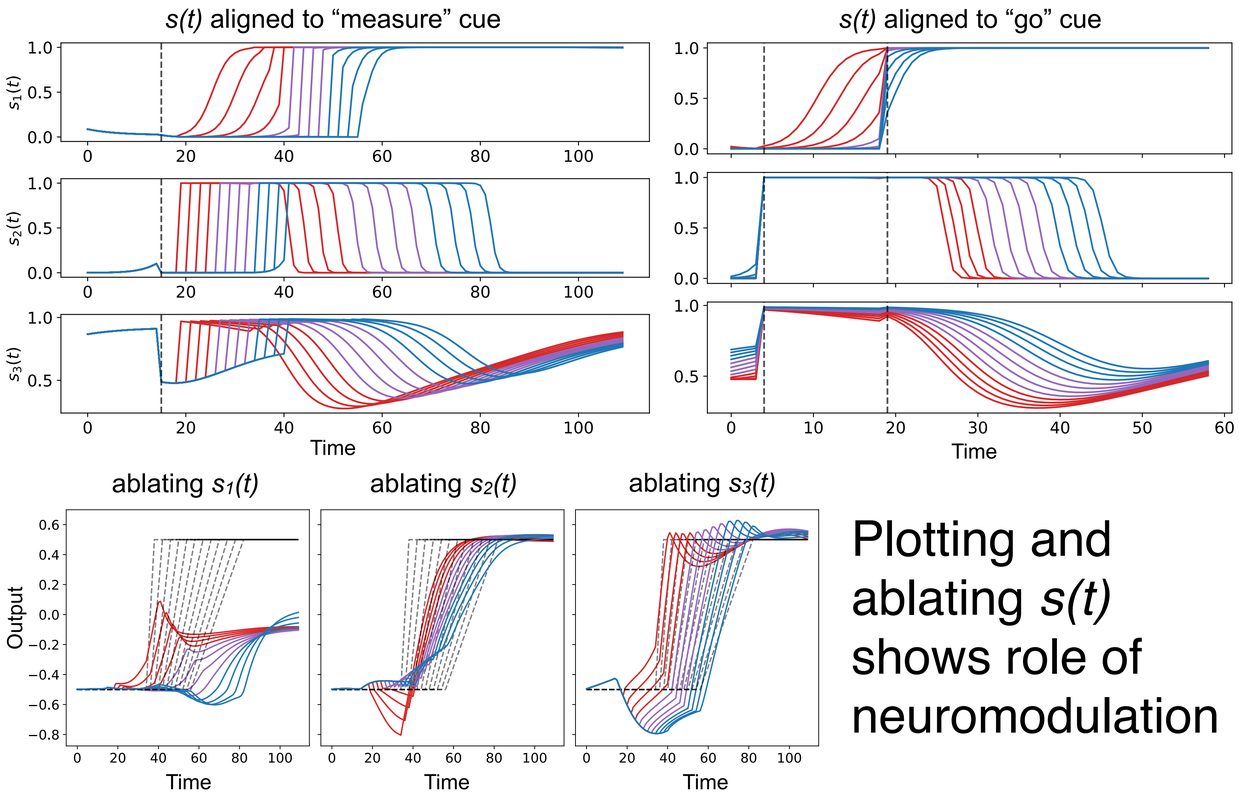
Primary task in [2], involves measuring, remembering, and reproducing time interval

Results

NM-RNN Improves Generalization on MWG



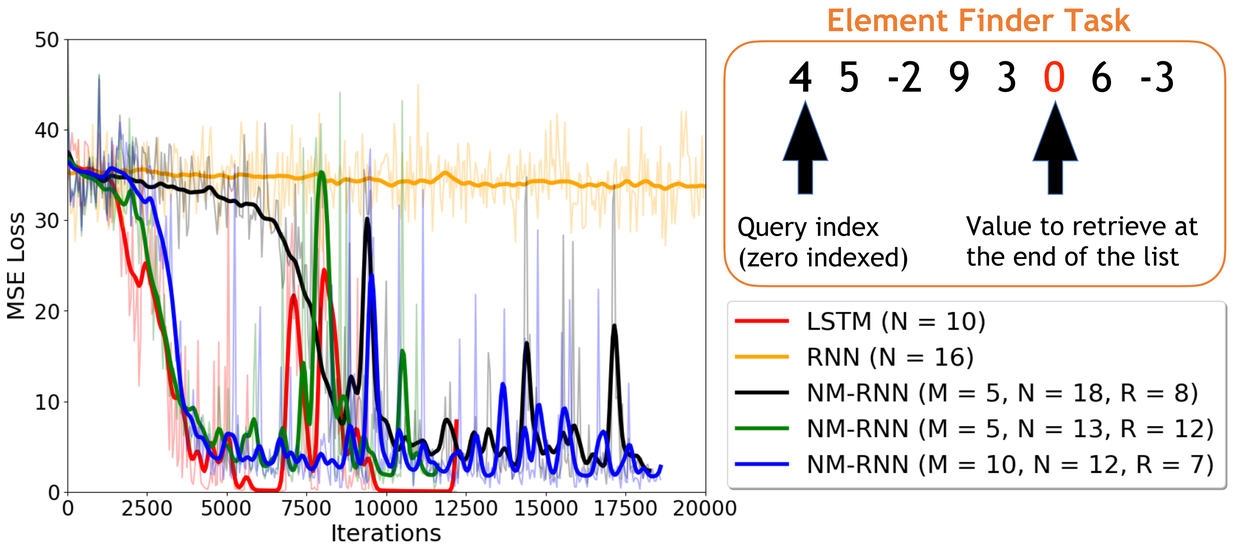
Rank-3 NM-RNN has improved performance over rank-3 RNN on trained intervals (purple) & extrapolation to unseen intervals (red, shorter; blue, longer)



Plotting and ablating *s(t)* shows role of neuromodulation

Connection to LSTMs

For a linear, symmetric-weight output network, neuromodulation acts like the forget gate of an LSTM. These networks can achieve LSTM-like performance on an "element finding" task that RNNs fail to learn.



Conclusions & Future Work

- NM-RNN adds interpretable neuromodulatory state to RNN models of neural circuits
- How might we integrate exciting neuromodulatory experimental data into our model?
- How might we gain a more exact characterization of the dynamical signatures induced by neuromodulation?

References

[1] Hamilos et al. (2021). "Slowly evolving dopaminergic activity modulates the moment-to-moment probability of reward-related self-timed movements." *eLife* 10:e62583
[2] Beiran et al. (2023). "Parametric control of flexible timing through low-dimensional neural manifolds." *Neuron 111 (5), 739-753. e8*